1. **Class of IP address 4.5.6.7:** a) Class A
2. **Loop-back address:** d) Host; itself
3. **Destination address for broadcasting:** c) 255.255.255.255
4. **Repeaters function in which OSI layer:** a) Physical
5. **Bridge accesses which address:** a) Physical
6. **As data packet moves from lower to upper layers, headers are:** a) Added
7. **Main function of the transport layer:** b) Process to process message delivery
8. **IPv6 address size:** c) 128
9. **Router looks at which address:** b) IP Address
10. **Application layer service:** d) All of the above
11. **Checksum in IP packet covers:** c) The header and data
12. **Not an IPC event:** d) Socket
13. **Socket uses:** a) Port number
14. **Difference between SOCK_STREAM and SOCK_SEQPACKET:** d) One is sequenced, the other is non-sequenced
15. **Advantage of RAW socket:** c) Direct interface to IP
16. **Data byte ordering in sockets:** c) Depends on hardware
17. **About bind():** d) As client does not listen to incoming request, bind is not necessary
18. **In TCP connection with sendto:** a) Returns error
19. **OSI layer for identifying communications partners:** a) Application
20. **Broadcast message dialog control method:** d) Simplex
21. **Router definition:** b) A network host that reads the source and destination addresses in the packet header and makes decisions about where to forward the packet
22. **Class C IP address:** d) 204.67.118.54
23. **Suitable for mission critical applications:** c) Token Ring
24. **Default subnet mask for Class C network:** c) 255.255.255.0
25. **Topology with each computer connected to a central point:** c) Star
26. **Most fault tolerant topology:** b) Mesh
27. **Bus network topology description:** a) All computers are attached to a single cable in a chain style
28. **Common cable type for 10BaseT Ethernet:** d) UTP
29. **Hub works with which topology:** b) Star
30. **Correct MAC address:** c) 08:34:FE:4C:F2:54
31. **Hub operates at which OSI layer:** a) Physical
32. **Devices operating at Data Link layer:** c) A Switch
33. **Router operates in which OSI layer:** d) The Network Layer
34. **Correct order of OSI model:** c) Physical, Data Link, Network, Transport, Session, Presentation, Application
35. **TCP/IP protocol for Web Browser and Server communication:** b) HTTP
36. **Bridge operates in which OSI layer:** a) The Data Link Layer
37. **Protocol for fast connectionless communications:** b) UDP
38. **Fastest method of transporting data with a switch:** b) Cut through
39. **Topology with highest redundancy:** c) Mesh
40. **OSI

You

with options

BLACKBOXAI

Sure! Here are the correct answers along with their options:

1. **Class of IP address 4.5.6.7:**
   a) Class A
   b) Class B
   c) Class C
   d) Class D
   Ans: a) Class A

2. **Loop-back address is used to send a packet from the ____ to ___ on the network:**
   a) Host; all other hosts
   b) Router; all other hosts
   c) Host; specific host
   d) Host; itself
   Ans: d) Host; itself

3. **What destination address can be used to send a packet from a host with IP address 188.1.1.1 to all hosts on the network?**
   a) 188.0.0.0
   b) 0.0.0.0
   c) 255.255.255.255
   d) 188.1.255.255
   Ans: c) 255.255.255.255

4. **Repeaters function in which layer of OSI?**
   a) Physical
   b) Data link
   c) Network
   d) a and b
   Ans: a) Physical

5. **A bridge can access to the ____ address of a station on the same network:**
   a) Physical
   b) Network
   c) Service access point
   d) All
   Ans: a) Physical

6. **As the data packet moves from lower to upper layers, headers are:**
   a) Added
   b) Subtracted
   c) Rearranged
   d) Modified
   Ans: a) Added

7. **What is the main function of the transport layer?**
   a) Node-to-node delivery

b) Process to process message delivery
c) Synchronization
d) Updating and maintenance of routing tables
Ans: b) Process to process message delivery

8. **IPv6 has ___ bit addresses:**
a) 32
b) 64
c) 128
d) Variable
Ans: c) 128

9. **When a host on network A sends a message to a host on network B, which address does the router look at?**
a) Port
b) IP Address
c) Physical
d) None of the above
Ans: b) IP Address

10. **Which of the following is an application layer service?**
a) Network virtual terminal
b) File transfer, access, and management
c) Mail service
d) All of the above
Ans: d) All of the above

11. **The checksum in the IP packet covers:**
a) Just the header
b) Just the data
c) The header and data
d) Just the source and destination address
Ans: c) The header and data

12. **Which of the following is not an IPC event?**
a) Semaphore
b) Message queue
c) Shared memory
d) Socket
Ans: d) Socket

13. **Socket is using:**
a) Port number
b) Interrupt number
c) Memory address
d) All
Ans: a) Port number

14. **Out of these which is a major difference between SOCK_STREAM and SOCK_SEQPACKET:**
a) One is used for TCP, other for UDP
b) Both are the same. No difference at all
c) One with fixed length
d) One is sequenced, other is non-sequenced
Ans: d) One is sequenced, other is non-sequenced

15. **What is the advantage of RAW socket:**
a) Easy to program
b) Fast accessing
c) Direct interface to IP
d) None of these
Ans: c) Direct interface to IP

16. **What kind of data byte ordering is followed in sockets:**
a) Little endian
b) Big endian
c) Depends on hardware
d) None of these
Ans: c) Depends on hardware

17. **Which of the following is true about bind():**
a) Bind is must in client process
b) Returns non-zero positive integer on success
c) Third argument in bind is a pointer
d) As client does not listen to incoming request, bind is not necessary
Ans: c) Third argument in bind is a pointer

18. **In a TCP connection if sendto is used with valid struct sockaddr pointer in 5th argument then a) Returns error
b) No problem, data can be passed
c) Converts the TCP socket to UDP
d) None of these
Ans: a) Returns error

19. **Which of the following OSI layers is responsible for identifying communications partners:**
a) Application
b) Session
c) Network
d) Presentation
Ans: a) Application

20. **A broadcast message is an example of which dialog control method:**
a) Half Duplex
b) Base band
c) Broad band

d) Simplex

Ans: d) Simplex

21. **What is a router:**

a) A hardware device that connects dissimilar networks, such as Cat 5 cabling and FDDI

b) A network host that reads the source and destination addresses in the packet header and makes decisions about where to forward the packet

c) A network host that can forward LAN-based email messages onto the Internet, after repackaging them into the SMTP format

d) A software system that can translate between dissimilar networks such as Ethernet and Token Ring

Ans: b) A network host that reads the source and destination addresses in the packet header and makes decisions about where to forward the packet

22. **Which of the following is Class C IP address:**

a) 10.10.14.118

b) 135.23.112.57

c) 191.200.199.199

d) 204.67.118.54

Ans: d) 204.67.118.54

23. **Which of the following is suitable for mission critical and time sensitive applications:**

a) 10BaseT

b) 10Base2

c) Token Ring

d) Mesh

Ans: c) Token Ring

24. **What is the default subnet mask for Class C network:**

a) 255.0.0.0

b) 255.255.0.0

c) 255.255.255.0

d) 255.0.255.0

Ans: c) 255.255.255.0

25. **Which of the following network topologies have each computer connected to a central point:**

a) Bus

b) Ring

c) Star

d) Mesh

Ans: c) Star

26. **Which of the following network topologies is the most fault tolerant:**

a) Bus

b) Mesh

c) Star

d) Ring

Ans: b) Mesh

27. **A Bus network topology is best described as which of the following:**

a) All computers are attached to a single cable in a chain style

b) All computers are connected to a central point

c) All computers are connected to every other computer or resource in the network

d) Fault Tolerant

Ans: a) All computers are attached to a single cable in a chain style

28. **10BaseT Ethernet networks, most commonly use which of the following types of cable:**

a) Fiber

b) Coax

c) STP

d) UTP

Ans: d) UTP

29. **A hub works in conjunction with which of the following network topologies:**

a) Ring

b) Star

c) Bus

d) Mesh

Ans: b) Star

30. **Which of the following is a correct MAC address:**

a) 190.168.80.1

b) www.certyourself.com

c) 08:34:FE:4C:F2:54

d) G2:G3:23:87:GG

Ans: c) 08:34:FE:4C:F2:54

31. **A Hub operates at which of the following layers of the OSI model:**

a) Physical

b) Session

c) Transport

d) Application

Ans: a) Physical

32. **Which of the following devices operate at the Data Link layer of the OSI model:**

a) A Hub

b) A NIC

c) A Switch

d) A Router

Ans: c) A Switch

33. **A Router operates in which layer of the OSI model:**

a) The Physical Layer

b) The Data Link Layer

c) The Transport Layer
d) The Network Layer
** Ans: d) The Network Layer**

34. **Which of the following is the correct order of the OSI model from bottom to top:**
a) Application, Presentation, Session, Transport, Network, Data Link, Physical
b) Physical, Network, Data Link, Session, Transport, Application, Presentation
c) Physical, Data Link, Network, Transport, Session, Presentation, Application
d) Application, Session, Presentation, Transport, Network, Data Link, Physical
Ans: c) Physical, Data Link, Network, Transport, Session, Presentation, Application

35. **Which of the following TCP/IP protocols is used to communicate between a Web Browser and a Web Server:**
a) SMTP
b) HTTP
c) UDP
d) POP3
Ans: b) HTTP

36. **Which of the following layers of the OSI model does a Bridge operate in:**
a) The Data Link Layer
b) The Network Layer
c) The Transport Layer
d) The Session Layer
Ans: a) The Data Link Layer

37. **Which of these protocols provide fast connectionless communications that relies on upper layers of the OSI model for error correction:**
a) TCP
b) UDP
c) SPX
d) None
Ans: b) UDP

38. **The fastest method of transporting data with a switch:**
a) Store and forward
b) Cut through
c) Loop back
d) None
Ans: b) Cut through

39. **Topology that offers the highest level of redundancy, is easy to troubleshoot but is expensive to install. It is the topology of the internet:**
a) Star
b) Ring
c) Mesh
d) Bus
Ans: c) Mesh

40. **OSI layer responsible for mail and file transfers:**
a) Transport
b) Data Link
c) Application
d) Physical
Ans: c) Application

41. **What is the first octet range for a class A IP address:**
a) 1 – 126
b) 1 – 128
c) 1 – 198
d) None
Ans: a) 1 – 126

42. **Network connection device that is capable of sending packets along multiple paths depending on which path is the most efficient:**
a) Bridge
b) Router
c) All the above
d) None of the above
Ans: b) Router

43. **OSI layer that determines whether the transmission is a full or half-duplex transfer:**
a) Presentation
b) Session
c) Application
d) Physical
Ans: d) Physical

44. **This OSI layer packages bits into frames:**
a) Physical
b) Data Link
c) Network
d) Application
Ans: b) Data Link

45. **OSI layer responsible for error checking frames and adding MAC addresses to its tables:**
a) Presentation
b) Transport
c) Data Link
d) Network
Ans: c) Data Link

46. **The initial formation of the packet header is started at this OSI level:**
a) Session
b) Application
c) Physical

d) Data Link

Ans: d) Data Link

47. **For a Class C IP address the first ___ octet(s) are reserved for the network ID:**

a) 1

b) 2

c) 3

d) 4

Ans: c) 3

48. **A set of "rules" for network transmissions:**

a) Topologies

b) Protocols

c) Gateways

d) All

Ans: b) Protocols

49. **What happens when a device is disconnected from the middle of a BUS network:**

a) The hub will propagate error messages to the rest of the network

b) All devices on the network will fail

c) The Server will stop responding to that device

d) Data will not be able to get to devices on the other side of this device

e) If a bridge is on the network, only that segment will be affected

Ans: d) Data will not be able to get to devices on the other side of this device

50. **IP is responsible for sorting and _____ of packets:**

a) Addressing

b) Holding

c) Organ d) Delivery

Ans: d) Delivery

51. **What is the first octet range for a class B IP address?**

a) 127-191

b) 128-191

c) 192-223

d) 1-127

Ans: b) 128-191

52. **What is the first octet range for a class C IP address?**

a) 1-126

b) 128-191

c) 192-223

d) 192-225

Ans: c) 192-223

53. **How many bits are there in an IP address V4?**

a) 16

b) 32

c) 64
d) 128
Ans: b) 32

54. **A network bridge operates at which layer of the OSI model?**
a) Physical
b) Data Link
c) Network
d) Transport
Ans: b) Data Link

55. **Each network interface card is uniquely identified by its?**
a) Transceiver type
b) MAC address
c) IP address
d) WINS address
Ans: b) MAC address

56. **The topology most commonly used with a token ring network is the?**
a) Star
b) Star-Bus
c) Ring
d) Star-Ring
Ans: c) Ring

57. **The primary purpose of a network is?**
a) Interconnect computers for ease of administration
b) Provide central access to workstation hard disks
c) Share resources
d) Provide increased security for all workstations
Ans: c) Share resources

58. **The recommended upper limit for the number of workstations that can be included in a peer-peer network is?**
a) 5
b) 10
c) 12
d) Unlimited
Ans: b) 10

59. **An ethernet network is a _____network?**
a) Active
b) Passive
c) Active-Passive
d) Passive-Aggressive
Ans: a) Active

60. **The OSI model layer that formats data packets and provides for error checking and correction is?**

a) Network
b) Data Link
c) Transport
d) Session
Ans: b) Data Link

61. **The network layer device that determines and sends packets are the most efficient route available is a _____?**
a) Bridge
b) Switch
c) Active hub
d) Router
Ans: d) Router

62. **Packets on the network layer are addressed using?**
a) Network address
b) Node address
c) MAC address
d) WINS address
Ans: a) Network address

63. **The process that enables faster devices to communicate with slower devices without the loss of data is?**
a) Error handling
b) Flow control
c) CRC
d) Parity
Ans: b) Flow control

64. **On the OSI model layers what is level 6?**
Ans: Presentation Layer

65. **On the OSI model what is level 7?**
Ans: Application Layer

66. **On the OSI model what is level 5?**
Ans: Session Layer

67. **On the OSI model what is level 4?**
Ans: Transport Layer

68. **On the OSI model what is level 3?**
Ans: Network Layer

69. **On the OSI model what is level 2?**
Ans: Data Link Layer

70. **On the OSI model what is level 1?**
Ans: Physical Layer

71. **What hardware devices does the data link layer deal with?**
Ans: Network Interface Cards (NICs), switches, bridges

72. **What hardware devices does the transport layer deal with?**
Ans: Hosts, computers, servers

73. **Segmentation of a data stream happens at which layer of OSI model?**
a) Physical
b) Data Link
c) Network
d) Transport
Ans: d) Transport

74. **When data is encapsulated, which is the correct order?**
a) data, frame, packet, segment, bit
b) segment, data, packet, frame, bit
c) data, segment, packet, frame, bit
d) data, segment, frame, packet, bit
Ans: c) data, segment, packet, frame, bit

75. **Which of the following would describe a transport layer connection that would ensure reliable delivery?**
a) Routing
b) Acknowledgment
c) Switching
d) System authentication
Ans: b) Acknowledgment

76. **What is a reason you want to use switches in your network instead of hubs?**
a) They're inexpensive
b) Switches are faster than hubs at reading frames
c) Switches create more collision domains
d) Switches do not forward broadcasts
Ans: b) Switches are faster than hubs at reading frames

77. **What protocol is used to find the hardware address of a local device?**
a) RARP
b) ARP
c) ICMP
d) IP
Ans: b) ARP

78. **What is the result of using a hierarchical addressing scheme?**
a) Increased number of addresses
b) Decreased number of routers needed
c) Increased memory usage on routers
d) No routing tables needed on routers
Ans: b) Decreased number of routers needed

79. **If you wanted to have 12 subnets using class C net ID, which subnet mask would you use?**
a) 255.255.255.252

b) 255.255.255.242
c) 255.255.255.240
d) 255.255.255.255
Ans: c) 255.255.255.240

80. **Which of the following statement is true?**
a) TCP is connection-oriented but doesn't use flow control
b) IP is not necessary on all hosts that use TCP
c) ICMP must be implemented by all TCP/IP hosts is False
d) ARP is used to find IP address from known hardware address is False
Ans: c) ICMP must be implemented by all TCP/IP hosts is False

81. **Which class of IP address provides a maximum of only 254 host addresses per network ID?**
a) Class A
b) Class B
c) Class C
d) Class D
Ans: c) Class C

82. **Which protocol uses ICMP?**
a) Telnet
b) Ping
c) ARP
d) IP
Ans: b) Ping

83. **Maximum port number allowed in socket is:**
a) Up to 4 GB
b) Up to 65535
c) No limit
d) 255
Ans: b) Up to 65535

84. **To use the UDP socket for communication:**
a) IP address is not essential
b) Port number is optional
c) a and b
d) None
Ans: a) IP address is not essential

85. **Accept system call returns:**
a) A new socket ID on success
b) Replaces the existing socket ID on failure
c) a and b
d) None
Ans: a) A new socket ID on success

86. **What is the default subnet mask of the IP 10.0.0.10?**

Ans: Class A

Explanation: The IP address 10.0.0.10 falls within the range of Class A addresses (1.0.0.0 to 126.0.0.0), the default subnet mask would be 255.0.0.0.

87. **Which of the following service defines the responsibility of the presentation layer?**

a) Format & code conversion
b) Establish, manage, and terminate session
c) Ensures reliability of transmission of data
d) Log-in & log-out procedures

Ans: a) Format & code conversion

88. **In LAN, using IEEE 802.3 Ethernet protocol, what is the topology followed?**

a) Can be either bus or star
b) Must be logical bus
c) Must be physical bus
d) Must be logical star

Ans: a) Can be either bus or star

89. **The destination address field in 802.3 frame is of _____bytes.**

a) 2
b) 4
c) 6
d) 8

Ans: c) 6

90. **A bridge used inside a LAN can read incoming:**

a) Frame
b) Datagram
c) Both
d) None

Ans: a) Frame

91. **A layer-2 switch works by using:**

a) MAC address
b) IP address
c) Subnet address
d) None

Ans: a) MAC address

92. **For a host to communicate in LAN/WAN, the IP address:**

a) Must be unique
b) Can be dummy
c) Is optional
d) Not required

Ans: a) Must be unique

93. **Which of the following is true about ARP?**

a) Request is broadcast but reply is unicast
b) Request is unicast but reply is broadcast
c) Request and reply both are unicast
d) Request and reply both are broadcast

Ans: a) Request is broadcast but reply is unicast

94. **The RARP packet in a host:**

a) Knows its physical address
b) Knows its IP address
c) a & b
d) None

Ans: b) Knows its IP address

95. **The maximum total size and maximum fragment size of IP datagram is:**

a) 65,535 bytes and depends on MTU by frame
b) 65,535 and 65,535
c) 65535 & 576
d) None of the above

Ans: c) 65535 & 576

96. **What is the maximum size of data that can be stuffed into UDP?**

a) 1024 bytes
b) 64KB
c) 512 bytes
d) Any size

Ans: d) Any size

97. **What are the port numbers of the following SMTP, TFTP, FTP?**

a) 21, 25, 69
b) 25, 69, 21
c) 69, 21, 25
d) None

Ans: b) 25, 69, 21

98. **What is the responsibility of TCP?**

a) Multiplexing
b) Reliability
c) Flow control
d) All

Ans: b) Reliability

99. **Which function is used to communicate data in connectionless mode?**

a) sendto
b) send
c) write
d) Any of the above

Ans: d) Any of the above

100. **htons() function returns:**

a) 16-bit integer in network byte order
b) 16-bit integer in host byte order
c) a or b
d) None

Ans: a) 16-bit integer in network byte order

101. **send() function returns:**

a) -1 on error
b) No. of bytes sent on success
c) a & b
d) None

Ans: c) a & b

102. **What is the second argument in 'connect()'?**

a) struct sockkaddr *
b) struct sockaddr
c) sizeof(struct sockaddr)
d) None

Ans: b) struct sockaddr

103. **When the incoming request queue created by a listen in a server is "listen(socket ID, 11)", then which of the following statement is true?**

a) If 10 active clients are communicating then the 11th client's connection request fails.
b) If 11 clients are actively connected then 12th client's connection request fails.
c) 100 users concurrently can use the server.
d) None of these.

Ans: b) If 11 clients are actively connected then 12th client's connection request fails.

104. **What is the default subnet mask for Class A network?**

a) 255.0.0.0
b) 255.255.0.0
c) 255.255.255.0
d) 255.0.255.0

Ans: a) 255.0.0.0

105. **UDP offers which of the following benefits relative to TCP?**

a) UDP consumes fewer computer resources by not maintaining connection state
b) UDP supports a self-regulating "throttle" feature that prevents network saturation
c) UDP guarantees that individual packets of a transmission will arrive "in order"
d) None of the above

Ans: a) UDP consumes fewer computer resources by not maintaining connection state

106. **TCP offers which of the following benefits relative to UDP?**

a) TCP allows much larger packets to be sent over the network, improving application performance
b) TCP includes a "retry" mechanism to prevent missing (dropped) packets
c) TCP uses a smaller header than does UDP

d) None of the above

Ans: b) TCP includes a "retry" mechanism to prevent missing (d ropped) packets

107. **The size of a UDP header is:**

a) 2 bytes
b) 4 bytes
c) 8 bytes
d) 16 bytes

Ans: c) 8 bytes

108. **In the current version of IP (IPv4), the use of TCP and UDP headers differ in which of the following ways?**

a) TCP checksums are optional but UDP checksums are mandatory
b) UDP checksums are optional but TCP checksums are mandatory
c) TCP checksums are larger than UDP checksums
d) UDP checksums are larger than TCP checksums

Ans: b) UDP checksums are optional but TCP checksums are mandatory

109. **The so-called "well-known" TCP and UDP port numbers fall into which numeric range?**

a) 0 to 999
b) 0 to 1000
c) 0 to 1023
d) 0 to 1024

Ans: c) 0 to 1023

110. **The so-called "dynamic" TCP and UDP port numbers fall into which numeric range?**

a) 8192 to 49151
b) 31768 to 54220
c) 49152 to 65535
d) 54220 to 65535

Ans: c) 49152 to 65535

111. **What is the size of the next version IP (IPv6)?**

a) 4 bytes
b) 8 bytes
c) 16 bytes
d) 32 bytes

Ans: c) 16 bytes

112. **Which of the following IPv4 addresses is the loopback address?**

a) 0.0.0.0
b) 10.0.0.1
c) 127.0.0.1
d) 255.255.255.255

Ans: c) 127.0.0.1

113. **The length of an IP datagram header can vary depending on the options used. What is the smallest possible IP header size?**

a) 16 bytes
b) 20 bytes
c) 24 bytes
d) 60 bytes

Ans: b) 20 bytes

114. **The source IP address for an IPv4 datagram is stored where in the header?**

a) 2nd longword (bytes 5-8)
b) 3rd longword (bytes 9-12)
c) 4th longword (bytes 13-16)
d) 5th longword (bytes 17-20)

Ans: a) 2nd longword (bytes 5-8)

115. **The datagram length field in an IPv4 datagram contains which numeric value?**

a) Total number of bytes in the datagram, including the header
b) Total number of bytes in the datagram, excluding the header
c) Total number of longwords in the datagram, including the header
d) Total number of longwords in the datagram, excluding the header

Ans: b) Total number of bytes in the datagram, excluding the header

116. **The maximum size of an IP datagram (including the header) is _____**

a) 1,024 bytes
b) 1,500 bytes
c) 4,096 bytes
d) 65,535 bytes

Ans: d) 65,535 bytes

117. **What function does Address Resolution Protocol (ARP) perform?**

a) Converts IP addresses to Ethernet addresses
b) Automatically assigns IP addresses to computers
c) Prevents two computers from using the same IP address
d) None of the above

Ans: a) Converts IP addresses to Ethernet addresses

118. **Which utility program is designed to report whether a networked computer is responding at a given IP address, and how long these responses take?**

a) Traceroute
b) Ping
c) TTCP
d) Netstat

Ans: b) Ping

119. **Which of these shell commands supplies the IP address of a computer running the Windows operating system?**

a) ipshow
b) traceroute
c) iplookup

d) ipconfig (or winipcfg)
Ans: d) ipconfig (or winipcfg)

120. **Each of the four numeric fields of an IP address written in dotted-decimal notation is also known as _____**
a) Octet
b) Word
c) Subnet
d) Class
Ans: a) Octet

121. **The Class D network range is reserved on the Internet for _____**
a) Broadcast
b) No cast
c) Anycast
d) None of the above
Ans: d) None of the above
Explanation: Class D network range is reserved for multicast addresses. Multicast allows a device to send a single packet to multiple destinations.

122. **A Class C network can handle a maximum of how many hosts?**
a) 254
b) 258
c) 510
d) 512
Ans: a) 254

123. **Which central authority is responsible for allocating IP addresses on the Internet?**
a) IETF
b) IANA
c) IEEE
d) None of the above
Ans: b) IANA

124. **Which Request For Comments (RFC) documentation describes the Internet Protocol?**
a) RFC 1
b) RFC 768 and RFC 1157
c) RFC 761
d) RFC 760 and RFC 791
Ans: d) RFC 760 and RFC 791

125. **The IP checksum feature supports integrity checking for _____**
a) An IP packet (header and payload)
b) An IP header only
c) An IP payload only
d) None of the above - IP doesn't use checksums
Ans: a) An IP packet (header and payload)

126. **The IP delivery service is _____**

a) Reliable, connection-oriented
b) Non-routable
c) Unreliable, connectionless
d) None of the above

Ans: c) Unreliable, connectionless

127. **MTU stands for _____**

a) Minimum Transfer Unit
b) Minimum Transmission Unit
c) Maximum Transmission Unit
d) Maximum Transfer Unit

Ans: c) Maximum Transmission Unit

128. **Which of these network devices belong at the OSI data link layer (layer two)?**

a) Router
b) Bridge
c) VPN
d) None of the above

Ans: b) Bridge

129. **Which of these network devices primarily functions at the OSI Network layer (layer 3)?**

a) Switch
b) Gateway
c) Router
d) All of the above

Ans: c) Router

130. **Which Protocol Data Unit (PDU) is employed at the Transport Layer?**

a) Bits
b) Frames
c) Packets
d) Segments

Ans: d) Segments

131. **What layer of the OSI model is associated with the MAC and LLC sublayers?**

a) Physical
b) Data Link
c) Network
d) Transport

Ans: b) Data Link

132. **What is out-of-band data?**

a) Urgent data
b) More data than bandwidth
c) Data in disorder
d) None of these

Ans: a) Urgent data

133. **If bind() fails on a socket what is the return?**

a) Nothing
b) 0
c) -1
d) None

Ans: c) -1

134. **After bind() failure, what is the best operation to complete the process?**

a) Call exit
b) Use return
c) Use close(), then close the process
d) Any of these

Ans: c) Use close(), then close the process

135. **Stream sockets are:**

a) Half duplex
b) Simplex
c) Full duplex
d) None

Ans: c) Full duplex

136. **Why is the bind() function not used on the client side?**

a) Because connect function serves the purpose
b) Can be used
c) To avoid extra connection
d) None

Ans: c) To avoid extra connection

137. **What is the loopback address for the system with IP 192.168.1.21?**

a) 192.168.1.255
b) 0.0.0.0
c) 127.0.0.1
d) None

Ans: c) 127.0.0.1

138. **What is the broadcast address for the 1st subnet, with IP 192.168.1.21 and mask as 255.255.255.192?**

a) 255.255.255.255
b) 255.255.255.0
c) 255.255.255.192
d) None

Ans: c) 255.255.255.192

139. **sendto() is used:**

a) In connection mode
b) In connection-less mode
c) Raw socket mode
d) None

Ans: b) In connection-less mode

140. **Packets in the IP layer are called _____.**
Ans: IP Packets or Datagrams

141. **Given the IP address 126.110.16.7 what class of address is it?**
Ans: Class A

142. **The error detection method used by IP is _____.**
Ans: Checksum

143. **Router works in how many layers? What are they?**
Ans: Routers work in three layers of the OSI model. They are:

144. Network Layer (Layer 3)

145. Data Link Layer (Layer 2)

146. Physical Layer (Layer 1)

147. **Which of the following statements is true?**
a) A switch supports more ports than a bridge.
b) In switches, each individual port could be configured for different data rates.
c) Both
d) None of this
Ans: c) Both

148. **What is the Broadcast address of a system with IP as 191.0.1.191?**
Ans: 191.0.1.255
Explanation: The broadcast address of a system with IP address as 191.0.1.191 with a subnet mask of 255.255.255.0 is 191.0.1.255.

149. **What is the subnet mask for an IP address 192.168.1.2 with 10 subnets?**
Ans: 255.255.240.0.
Explanation: To create 10 subnets, borrow 4 bits from the host portion, making the subnet mask: 255.255.240.0.

150. **Ethernet is also known as?**
a) 802.5
b) 803.2
c) 802.11
d) 802.3
Ans: d) 802.3

151. **In a Basic Ring topology, if there is a line break, or if a user is adding or removing a device anywhere in the ring, this will bring down the network. (True / False).**
Ans: True

152. **Mesh topologies are the least expensive and are easier to maintain. (True / False).**
Ans: False

153. **If computers are connected along a single cable, this is called what type of topology?**
Ans: Bus Topology

154. **The number of computers on a bus network will NOT affect network performance. (True / False).**
Ans: False
Explanation: The number of computers on a bus network affects network performance because all data transmitted on the bus is received by every device, leading to potential congestion and collisions.

155. **What is the size of an IP fragment?**
Ans: The size of an IP fragment depends on the Maximum Transmission Unit (MTU) of the network over which it is transmitted.

156. **How many bytes is the sequence number field in TCP Header?**
Ans: 4 bytes or 32 bits

157. **ARP implemented in a switch. (TRUE / FALSE).**
Ans: FALSE
Explanation: ARP (Address Resolution Protocol) is not implemented in a switch. It is typically implemented in hosts and routers to map IP addresses to MAC addresses.

158. **IP fragment carries IP Header. (TRUE/ FALSE).**
Ans: FALSE
Explanation: IP fragments carry the IP payload (data) along with the IP header.

159. **List out the flags used in TCP Header.**
Ans: The flags used in the TCP header are:

160. URG (Urgent)

161. ACK (Acknowledgment)

162. PSH (Push)

163. RST (Reset)

164. SYN (Synchronize)

165. FIN (Finish)

166. **What is the return value of gethostbyname() on success?**
Ans: Pointer to a hostent structure.

167. **For a Class B IP address, the first ___ octet(s) are reserved for the network ID?**
Ans: 2 octets

168. **What is the address of the limited Broadcast address?**
Ans: 255.255.255.255

169. **Which Protocol Data Unit (PDU) is employed at the Data Link Layer?**
Ans: Frame

170. **Which Protocol Data Unit (PDU) is employed at the Network Layer?**
Ans: Packet

171. **Which Protocol Data Unit (PDU) is employed at the Physical Layer **Ans: Bit**

172. **What are the smallest and largest possible values for an IP octet?**
Ans: 0 and 255

173. **How to convert a string format of IP address into a protocol-specific internet address?**
Ans: inet_pton() function from the arpa/inet.h header file.

174. **Which of the following statements is(are) true?**
a) TFTP uses UDP services
b) TFTP has a flow-control mechanism
c) TFTP has five types of messages
d) All of the above
Ans: a) TFTP uses UDP services
Explanation: TFTP (Trivial File Transfer Protocol) uses UDP services, does not have a flow-control mechanism, and has five types of messages.

175. **A bridge filters traffic using which type of address?**
Ans: MAC (Media Access Control) address.

176. **What topology provides each device with a point-to-point connection to every other device in the network?**
Ans: Mesh Topology

177. **What must be configured in TCP/IP for two computers on the same segment to communicate with each other?**
a) IP address and subnet mask
b) DNS address
c) All of the above
d) None
Ans: a) IP address and subnet mask

178. **For a Class A IP address, the first ___ octet(s) are reserved for the network ID?**
a) 1
b) 2
c) 3
d) 4
Ans: a) 1

179. **What is the subnet broadcast address for IP 192.168.1.100 with subnet mask as 255.255.255.224?**
Ans: 192.168.1.127.
Explanation: The broadcast address is calculated by flipping all host bits in the network portion of the network address to 1.

180. **When data is transmitted from device A to device B, the header from A's layer 5 is read by B's _____ layer.**

a) Physical
b) Transport
c) Session
d) Presentation
Ans: b) Transport

181. **Session layer checkpoints _____.**
a) Allow just a portion of a file to be resent
b) Detect and recover errors
c) Control the addition of headers
d) Are involved in dialog control
Ans: d) Are involved in dialog control

182. **Encryption and decryption are functions of the _____ layer.**
a) Transport
b) Session
c) Presentation
d) Application
Ans: c) Presentation

183. **Match the following to one of the seven OSI layers:**
a) Reliable end-to-end data transportation
Ans: Transport Layer
b) Network selection
Ans: Network Layer
c) Defines frames
Ans: Data Link Layer
d) Provides user services such as e-mail and file transfer
Ans: Application Layer
e) Transmission of bit stream across physical medium
Ans: Physical Layer

184. **A Packet from an Ethernet network requires a _____ before it can be routed to an FDDI network.**
a) Repeater
b) Bridge
c) Router
d) Gateway
Ans: c) Router

185. **Identify the following IP address: 169.5.0.0.**
a) Host IP address
b) Direct broadcast address
c) Limited broadcast address
d) Network address
Ans: d) Network address.

186. **Using the direct broadcast address, a_____ sends a packet to _____ on the network.**

a) Host; all other hosts
b) Router; all other hosts
c) Host; specific host
d) Host; itself
Ans: a) Host; all other hosts.

187. **Using the limited broadcast address, a_____ sends a packet to _____ on the network.**
a) Host; all other hosts
b) Router; all other hosts
c) Host; specific host
d) Host; itself
Ans: a) Host; all other hosts.

188. **Find the class of the following IP addresses:**
a) 208.34.54.12
Ans: Class C
b) 238.34.2.1
Ans: Class D
c) 114.34.2.8
Ans: Class A
d) 129.14.6.8
Ans: Class B
e) 241.34.2.8
Ans: Class E

189. **Find the netID and host ID of the following IP addresses:**
a) 192.8.56.2
Ans: NetID – 192.8.56, HostID - 2
b) 220.34.8.9
Ans: NetID – 220.34.8, HostID – 9
c) 208.34.54.12
Ans: NetID – 208.34.54, HostID – 12
d) 205.23.67.8
Ans: NetID – 205.23.67, HostID – 8

190. **A device has the IP address 190.1.2.3. What is its subnet ID?**
Ans: d) Insufficient information

191. **Which of the following is the default mask for the address 198.0.46.201?**
a) 255.0.0.0
b) 255.255.0.0
c) 255.255.255.0
d) 255.255.255.255
Ans: c) 255.255.255.0

192. **What class of IP address does the subnet mask 255.128.0.0 operate on?**
a) A
b) B

c) C
d) A or B or C
Ans: a) A
Explanation: Class A addresses have their first octet in the range 1-126, with a default subnet mask of 255.0.0.0.

193. **A subnet mask of 255.255.255.248 allows how many host IDs per subnet? (Disregard special addresses).**
a) 3
b) 6
c) 8
d) 248
Ans: b) 6
Explanation: Number of Hosts = 2^(Number of Host Bits) − 2 = 2^(3) - 2 = 8 - 2 = 6.

194. **A subnet mask of 255.255.255.224 allows how many host IDs per subnet? (Disregard special addresses).**
a) 3
b) 7
c) 8
d) 30
Ans: d) 30
Explanation: Number of Host Bits = 32 - 27 = 5. Number of Hosts = 2^(Number of Host Bits) − 2 = 2^(5) - 2 = 32 - 2 = 30.

195. **The subnet mask for a class C network is 255.255.255.192. How many subnetworks are available? (Disregard special addresses).**
a) 2
b) 4
c) 8
d) 192
Ans: b) 4
Explanation: 2^2 = 4 subnets.

196. **The subnet mask for a class B network is 255.255.255.192. How many subnetworks are available? (Disregard special addresses).**
a) 2
b) 4
c) 1,024
d) 192
Ans: c) 1,024
Explanation: The default subnet mask for Class B is 255.255.0.0. (/16) to make 255.255.255.192. (/26) Number of bits borrowed for subnetting = 10 (26-16). Number of subnets = 2^(number of bits borrowed for subnetting) = 2^(10) = 1024.

197. **The IP address of a host on a class C network is 198.123.46.237. Four subnets are allowed for this network. What is the subnet mask? Assume a contiguous subnet address space.**

Ans: 255.255.255.192.
Explanation: To accommodate four subnets, need to borrow at least two bits (2^2 = 4 subnets). By borrowing two bits for subnetting, changing the default subnet mask from 255.255.255.0 (/24) to 255.255.255.192 (/26).

198. **What is the subnetwork address and subnet mask for the IP address 142.45.43.44? Assume a contiguous subnet address space.**
Ans: 142.45.43.0 and 255.255.255.192.
Explanation: Assume to create 4 subnets, need to borrow 2 bits from the host portion to create the subnet portion (2^2 = 4). With 2 bits borrowed for subnetting, the new subnet mask will be 255.255.255.192. The subnetwork address is calculated as follows:
IP address: 10001110.00101101.00101011.00101100 (142.45.43.44)
Subnet mask: 11111111.11111111.11111111.11000000 (255.255.255.192)
Subnetwork address: 10001110.00101101.00101011.00000000 (142.45.43.0).

199. **What is the range of hosts for the subnetwork with an address of 150.201.93.4? Assume a contiguous subnet address space.**
Ans: The range of hosts is 0-255.
Explanation: The range of hosts depends on the subnet mask used, but typically for a /24 subnet, the range would be from 150.201.93.1 to 150.201.93.254.

200. **Find the class of the following subnetworks:**
a) 22.14.0.0
Ans: Class A
b) 133.45.6.0
Ans: Class B
c) 16.14.9.0
Ans: Class B
d) 193.45.127.64
Ans: Class C

201. **Find the subnetwork address for the following:**
a) IP Address: 125.34.12.56
Ans: 125.34.0.0
Mask: 255.255.0.0
b) IP Address: 120.14.22.16
Ans: 120.14.0.0
Mask: 255.255.128.0
c) IP Address: 140.11.36.22
Ans: 140.11.36.0
Mask: 255.255.255.0
d) IP Address: 141.181.14.16
Ans: 141.181.0.0
Mask: 255.255.224.0
e) IP Address: 200.34.22.156
Ans: 200.34.22.144
Mask: 255.255.255.240

Explanation: The subnetwork address is calculated by performing a bitwise AND operation between the IP address and the subnet mask.

202. **In _____ delivery, both the deliverer of the IP packet and the destination are on the same network.**
a) A connectionless
b) A connection-oriented
c) A direct
d) An indirect
Ans: c) A direct

203. **In _____ delivery, both the deliverer of the IP packet and the destination are on different networks.**
a) A connectionless
b) A connection-oriented
c) A direct
d) An indirect
Ans: d) An indirect

204. **In _____ delivery, packets of a message are logically connected to one another.**
Ans: Connection-oriented

205. **A best effort delivery service such as IP does not include _____.**
a) Error checking
b) Error correction
c) Datagram acknowledgement
d) All of the above
Ans: b) Error correction

206. **What is the value of the total length field if the header is 28 bytes and the data field is 400 bytes?**
a) 428
b) 407
c) 107
d) 427
Ans: a) 428
Explanation: Total length = Header size + Data field size = 28 bytes + 400 bytes = 428 bytes.

207. **The _____ module takes fragments of a message and puts them back in order.**
Ans: d) Reassembly

208. **Which fields of the IP header change from router to router?**
Ans: TTL (Time To Live), Checksum

209. **What is the first octet range for a class E IP address?**
Ans: 240 - 255

210. **Which protocol does the fragmentation?**

Ans: IP (Internet Protocol)